Práctica 1: Diseño de controlador

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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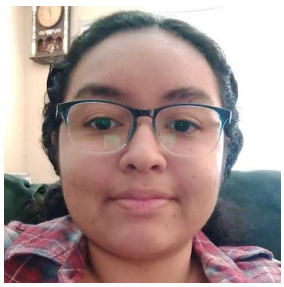
Información general



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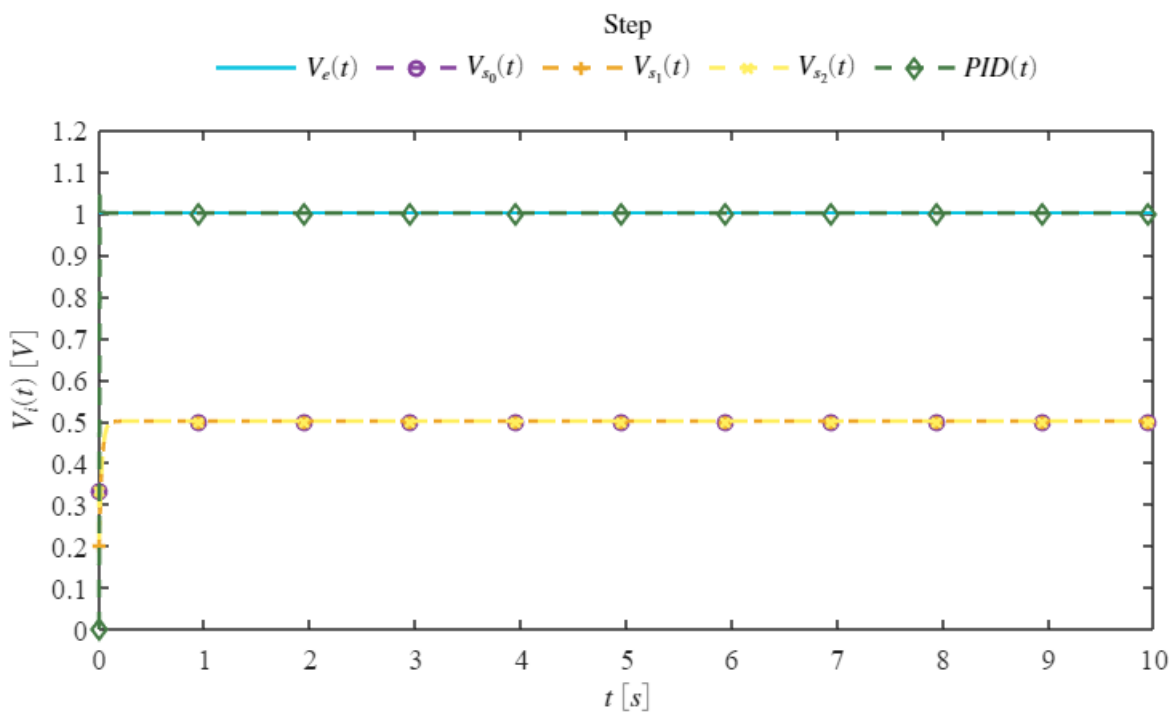
Datos de la simulación

```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sistema';
open_system(file);
parameter.StopTime = tend;
parameter.Solver = 'ode15s';
parameter.MaxStep = '1E-3';
%set_param('Sistema/Pao(t)','VectorFormat','1-D array');
```

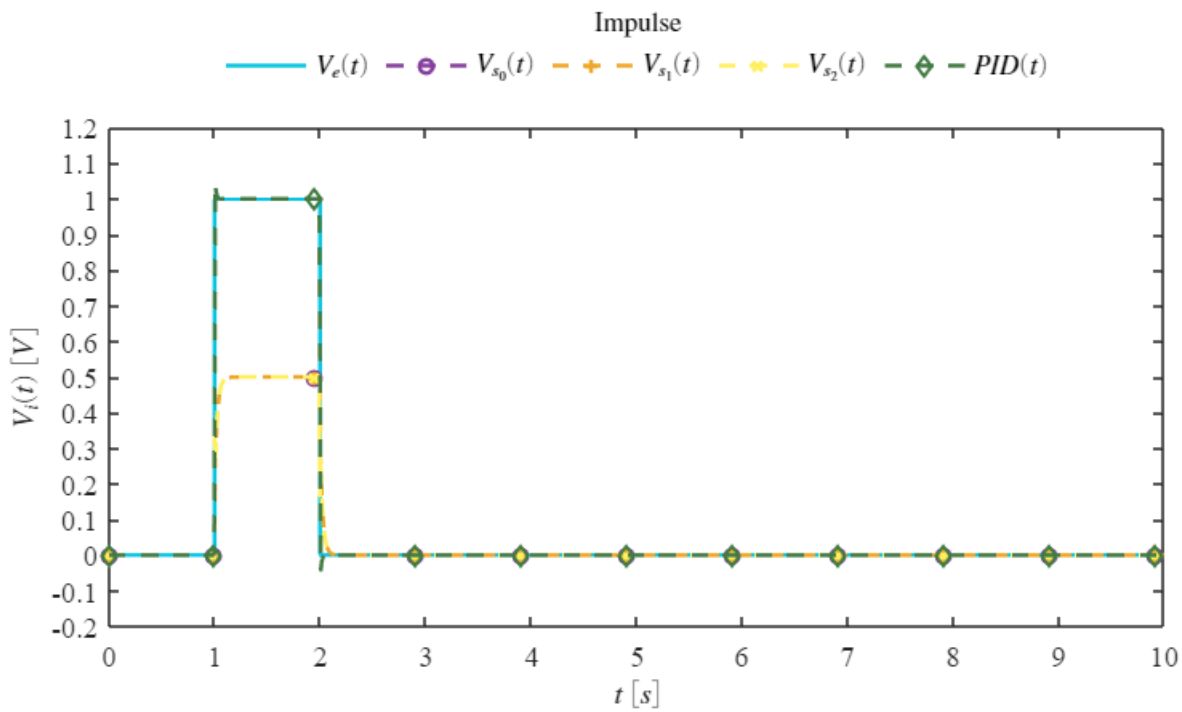
Respuesta al escalón

```
Signal ='Step';
set_param('Sistema/S1','sw','1');
set_param('Sistema/Ve(t)','sw','1');
x1 = sim(file,parameter);
plotsignals(x1.t,x1.Ve,x1.Vs0,x1.Vs1,x1.Vs2,x1.PID,Signal)
```



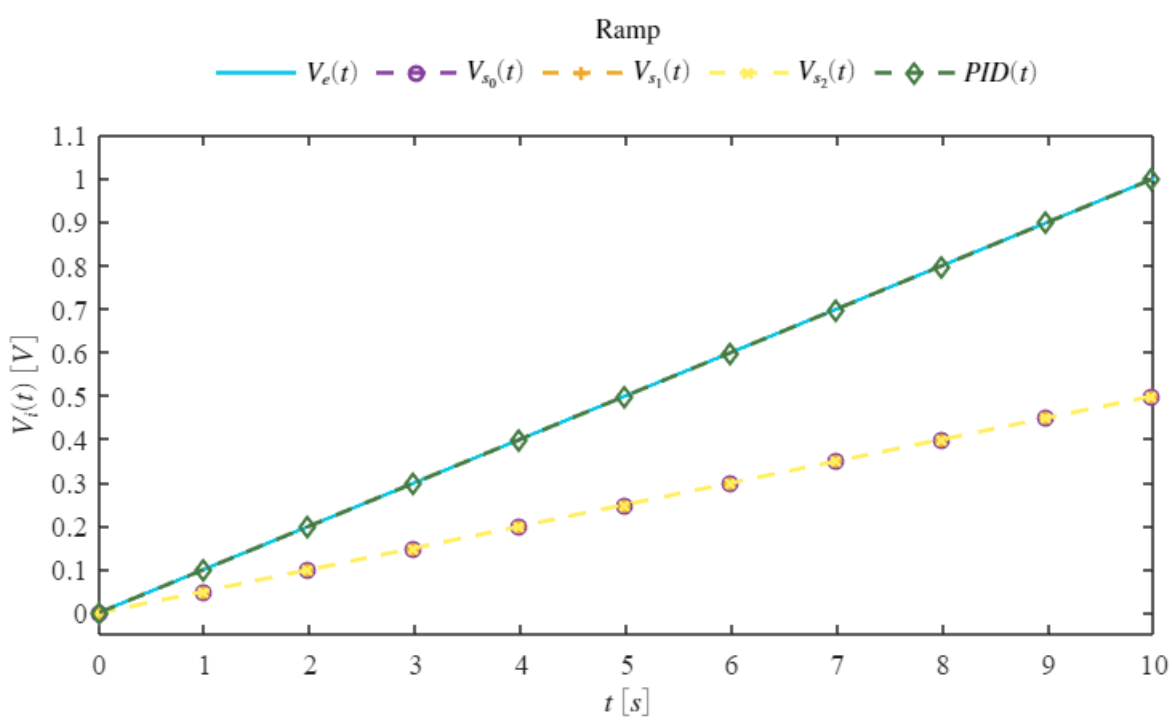
Respuesta al impulso

```
Signal ='Impulse';
set_param('Sistema/S1','sw','0');
set_param('Sistema/Ve(t)','sw','1');
x2 = sim(file,parameter);
plotsignals(x2.t,x2.Ve,x2.Vs0,x2.Vs1,x2.Vs2,x2.PID,Signal)
```



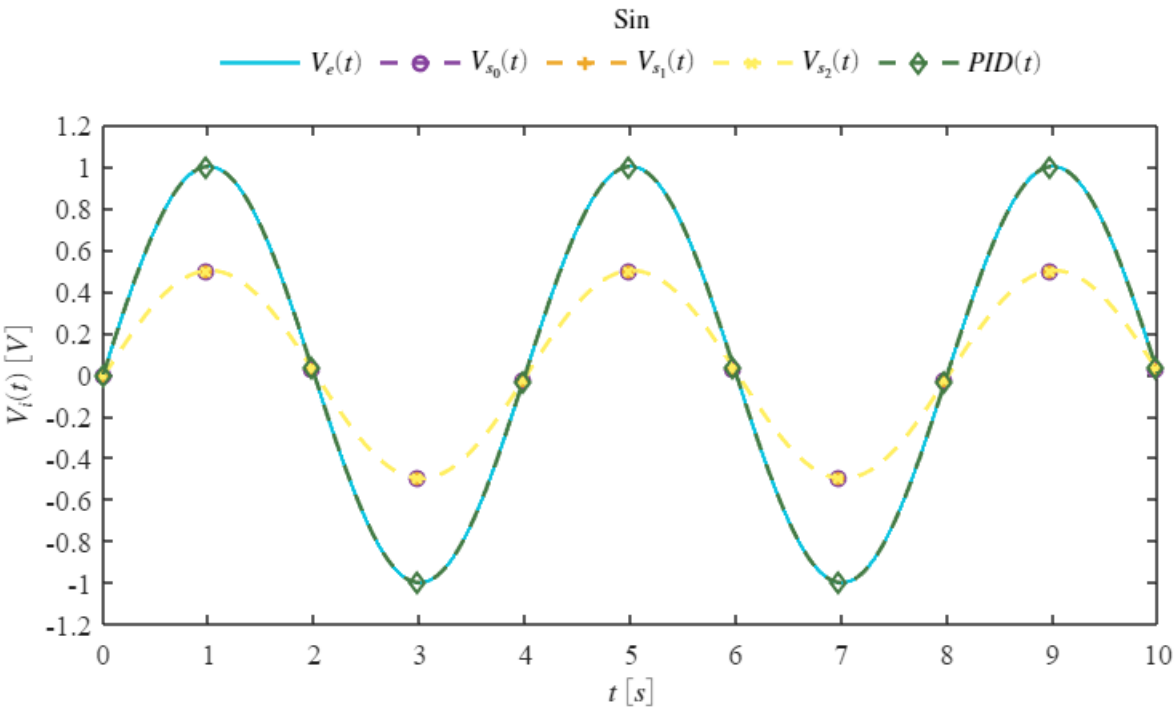
Respuesta a la rampa

```
Signal ='Ramp';
set_param('Sistema/S2','sw','1');
set_param('Sistema/Ve(t)','sw','0');
x3 = sim(file,parameter);
plotsignals(x3.t,x3.Ve,x3.Vs0,x3.Vs1,x3.Vs2,x3.PID,Signal)
```



Respuesta a la función sinusoidal

```
Signal ='Sin';
set_param('Sistema/S2','sw','0');
set_param('Sistema/Ve(t)','sw','0');
x4 = sim(file,parameter);
plotsignals(x4.t,x4.Ve,x4.Vs0,x4.Vs1,x4.Vs2,x4.PID,Signal)
```



Funcion: Respuesta a las senales

```
function plotsignals(t, Ve,Vs0,Vs1,Vs2,PID,Signal)
    set(figure(),'Color','w')
    set(gcf,'units','centimeters','position',[1,1,18,10])
    set(gca,'FontName','Times New Roman','FontSize',11)
    hold on; grid off; box on

    powerrangers = [10,196,224;
                    133,64,157;
                    238,167,39;
                    255,239,95;
                    66,122,67]/255;
    colororder(powerrangers)

    plot(t,Ve,'-',t,Vs0,'--o',t,Vs1,'--+',t,Vs2,'--x',t,PID,'--d',...
        'LineWidth',1.5,'MarkerSize',5,'MarkerIndices',1:1000:length(t));
    L = legend('$V_{e}(t)$','$V_{s_0}(t)$','$V_{s_1}(t)$','$V_{s_2}(t)$','$PID(t)$');
    set(L,'Interpreter','Latex','FontSize',10,'location',...
        'NorthOutside','box','off','Orientation','Horizontal')
    title(L,Signal,'FontSize',10)
```

```
    xlabel('$t$ $[s]$','Interpreter','Latex','FontSize',11)
    ylabel('$V_i(t)$ $[V]$','Interpreter','Latex','FontSize',11)

    if Signal == "Step"
        xlim([0,10]); xticks(0:1:10)
        ylim([0,1.2]); yticks(0:0.1:1.2)
    elseif Signal == "Impulse"
        xlim([0,10]); xticks(0:1:10)
        ylim([-0.2,1.2]); yticks(-0.2:0.1:1.2)
    elseif Signal == "Ramp"
        xlim([0,10]); xticks(0:1:10)
        ylim([-0.05,1.1]); yticks(0:0.1:1.2)
    elseif Signal == "Sin"
        xlim([0,10]); xticks(0:1:10)
        ylim([-1.2,1.2]); yticks(-1.2:0.2:1.2)
    end

    exportgraphics(gcf,[Signal,'.pdf'],'ContentType','vector')
    %exportgraphics(gcf,[Signal,'.png'],'Resolution',600);
    %print(Signal,'-dsvg','-r600');
    %print(Signal,'-depsc','-r600')

end
```